MAULANA AZAD NATIONAL INSTITUTEOF TECHNOLOGY, BHOPAL

NETWORK LAB

EXPERIMENT NO.04

Aim: To study & verify Maximum power transfer theorem.

Apparatus Required:

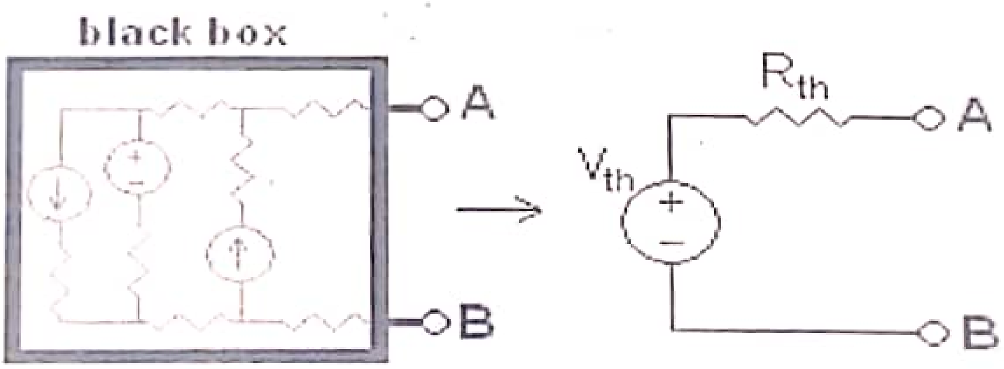
1. Trainer kit
2. Connecting cords

Theory:

Theorem: First of all, Maximum power transfer theorem states that:

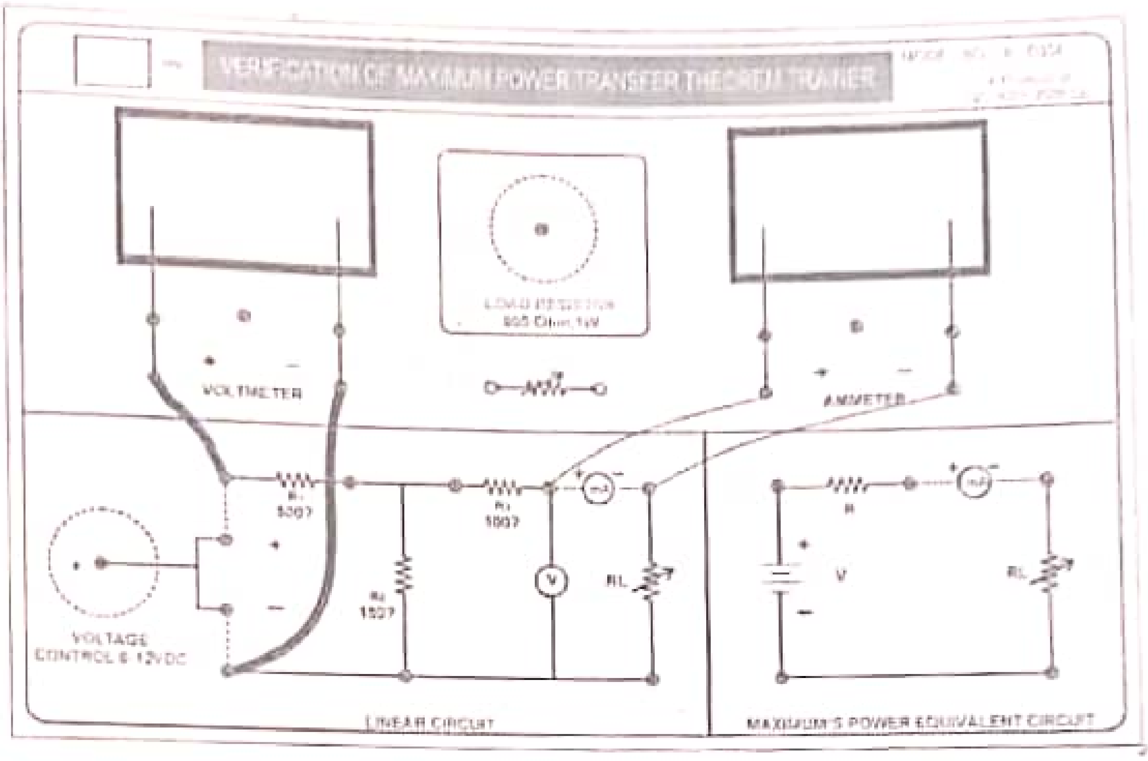
- Any linear electrical network with voltage and current only resistances can be replaced at terminals A-B by an equivalent sources and voltage source Vth in series connection with an equivalent resistance Rth. This equivalent voltage Vth is the voltage obtained at terminals A-B of the network with terminals A-B open circuited.

- This equivalent resistance Rth is the resistance obtained at terminals A-B of the network with all its independent current sources open circuited and all its independent voltage sources short circuited.

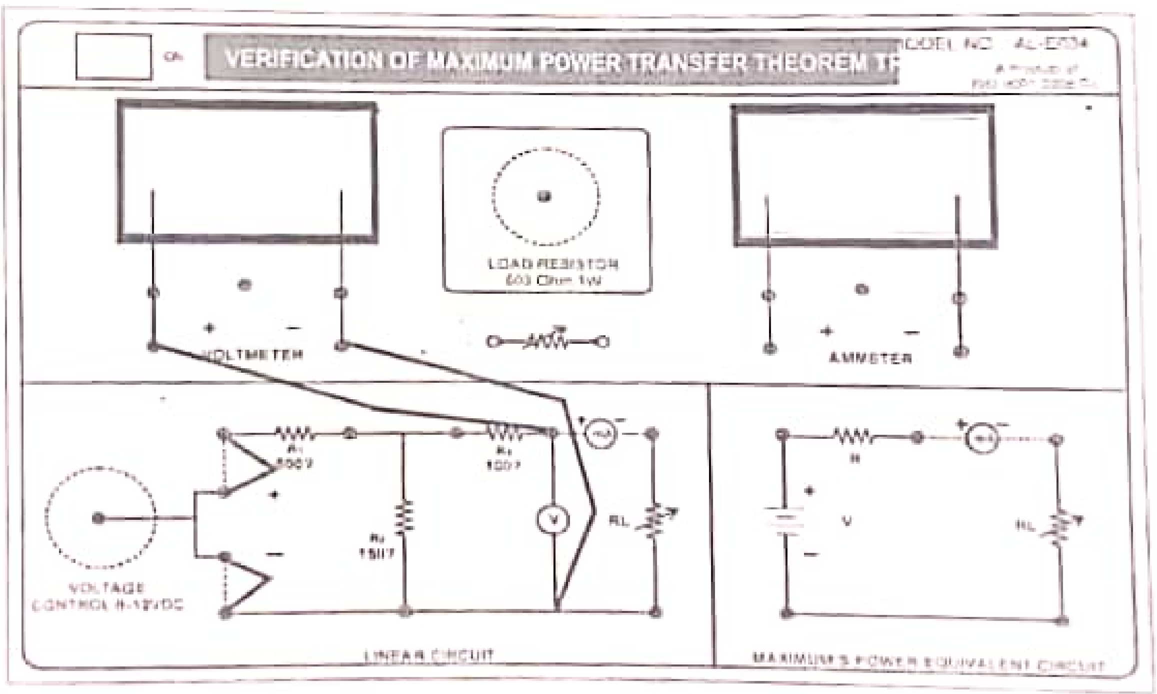


Connection diagram:

1. *For calculation of open circuit voltage (Maximum power transfer voltage)*



2. *For calculation of equivalent resistance (RTH) by Multi meter*



Any black box containing resistances only and voltage and current sources can be replaced to a Maximum power transfer equivalent circuit consisting of an equivalent voltage source in series connection with an equivalent resistance. We can verify maximum power transfer theorem with the help of Maximum power transfer theorem.

The power transferred from a supply source to a load is at its maximum when the resistance of the load is equal to the internal resistance of the source. On the other words" A resistive load will be consumptive maximum power from the supply when the load resister is equal to the equivalent (Maximum power transfer) **network resister"**

Procedure:

1. Connections should be as per the circuit diagram.

2. Short circuit the voltage source and open circuit the output terminal for calculation of equivalent resistance.

3. Connect the supply voltage and open the output terminal and we get Maximum power transfer voltage (Vth).

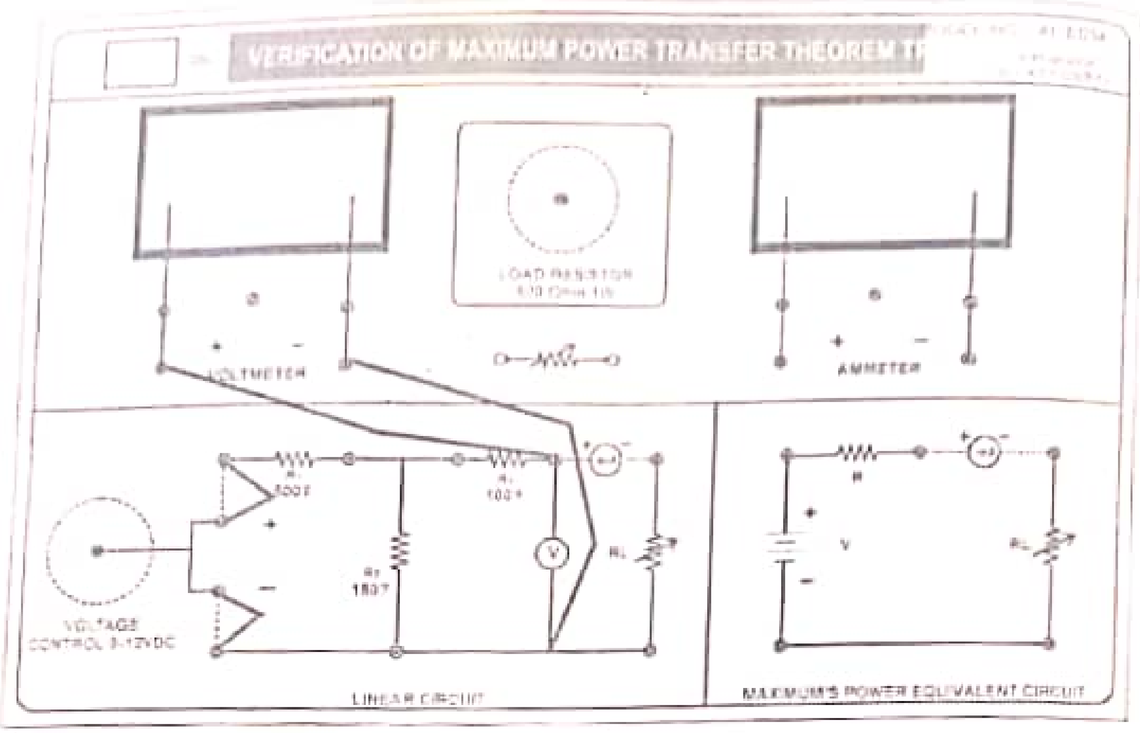
4. Now make a equivalent circuit with load connected and calculate the voltage across load (RL).

8. Using formula find out the Maximum power transfer voltage (Vth) and equivalent resistance (Req).

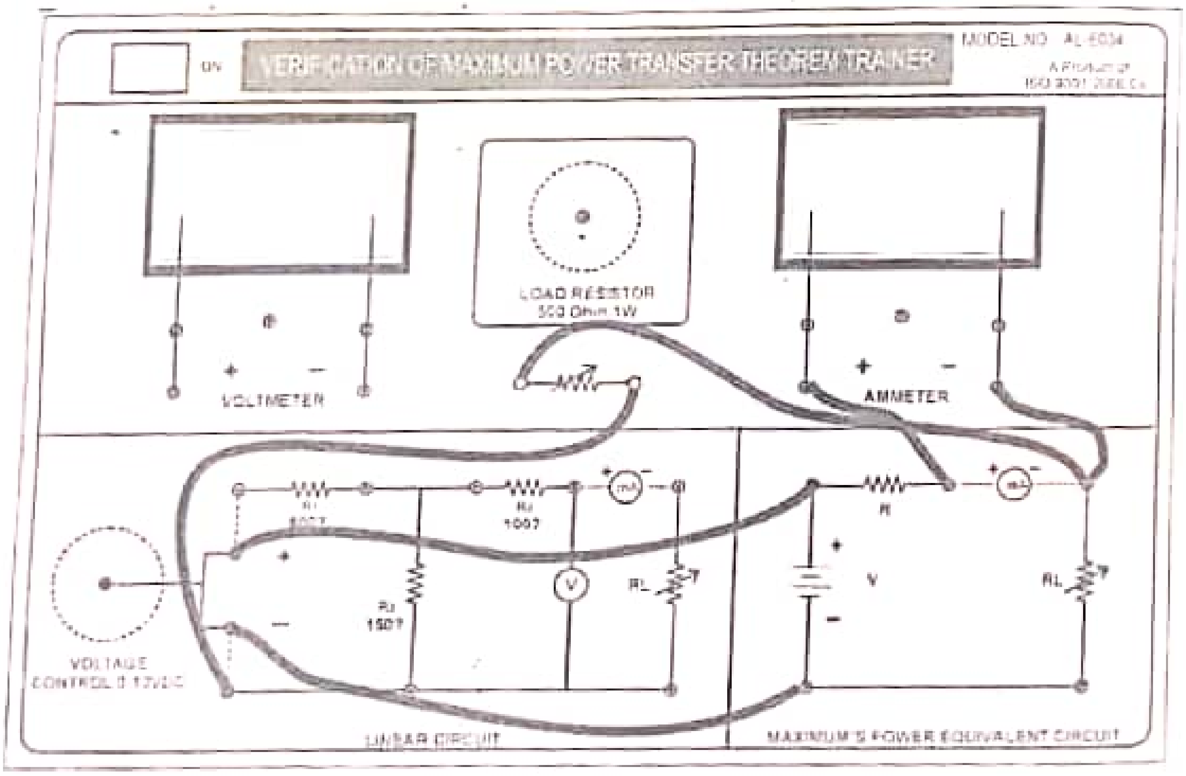
9. see here by the current at the resistance equal to the Maximum power transfer equivalent resistance have maximum, other will be reduced and increased depends upon the resistance.

10. Verify that power is maximum when RL =Rl.

3. Calculation of VTH



4. Maximum power transfer equivalent circuit with load



Observations and calculation:

Sr no.	Vin	Vth (Maximum power transfer voltage)		Req (equivalent resistance)	RL (Load resistance)	
		Practical	Theoretical		Practical	Theoretical
1.						
2.						
3.						
4.						
5.						

Result: When maximum power is transferred near RL=Req. Then theorem is verified.

Questions

1. What is the Maximum Power Transfer Theorem? Explain with an example
2. Does the Maximum Power Transfer Theorem apply to AC circuits?
3. What is the Maximum Power Transfer Formula?
4. What is Thevenin's Resistance?
5. What are the limitations of the Maximum Power Transfer Theorem?

Ans

1. Maximum Power Transfer Theorem explains that to generate maximum external power through a finite internal resistance (DC network), the resistance of the given load must be equal to the resistance of the available source.
   Large sound systems are built around this process. Maximum power transfer is generated in the circuit by making the speaker's (load) resistance equivalent to the resistance of the amplifier.
   Once the speaker and amplifier have equal resistance, both are considered harmonised.
2. Just like DC circuits, AC circuits work under the Maximum Power Transfer Theorem. But in contrast to DC circuits, resistance is swapped with impedance in AC electric circuits.
3. When $R_{th}=R_L$

   Then

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

4. Thevenin's Resistance is the resistance calculated at the given terminals with every voltage source reinstated with short circuits, and every current source replaced by open circuits.
5. The critical limitation of the Maximum Power Transfer Theorem is, it cannot be used in nonlinear and unilateral networks. As efficiency drops to 50%, it is also not applicable in power systems.